

3.6 First Passage Time Distribution

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Introduction

- In this section, we work only with Brownian motion, the continuous-time counterpart of the symmetric random walk.
- We begin here with a martingale containing Brownian motion in the exponential function.
- We fix a constant σ . The so-called *exponential martingale* corresponding to σ , which is

$$Z(t) = \exp \left\{ \sigma W(t) - \frac{1}{2} \sigma^2 t \right\}$$

Theorem 3.6.1 (Exponential martingale)

- Let $W(t)$, $t \geq 0$, be a Brownian motion with a filtration $F(t)$, $t \geq 0$, and let σ be a constant. The process $Z(t)$, $t \geq 0$, is a martingale.

$$Z(t) = \exp \left\{ \sigma W(t) - \frac{1}{2} \sigma^2 t \right\}$$

Proof of Theorem 3.6.1

- For $0 \leq s \leq t$, we have

$$\begin{aligned} E[Z(t)|F(s)] &= E\left\{\exp\left\{\sigma W(t) - \frac{1}{2}\sigma^2 t\right\} \middle| F(s)\right\} \\ &= E\left\{\exp\left\{\sigma W(t) - \sigma W(s) + \sigma W(s) - \frac{1}{2}\sigma^2 t\right\} \middle| F(s)\right\} \\ &= E\left\{\exp\{\sigma(W(t) - W(s))\} \cdot \exp\left\{\sigma W(s) - \frac{1}{2}\sigma^2 t\right\} \middle| F(s)\right\} \\ &= \exp\left\{\sigma W(s) - \frac{1}{2}\sigma^2 t\right\} \cdot E\{\exp\{\sigma(W(t) - W(s))\} | F(s)\} \\ &= \exp\left\{\sigma W(s) - \frac{1}{2}\sigma^2 t\right\} \cdot E[\exp\{\sigma(W(t) - W(s))\}] \end{aligned}$$

$E[XY|g] = XE[Y|g]$
X is g-msb

$E[X|g] = EX$
X is independent of g

Proof of Theorem 3.6.1 (2)

- $W(t)-W(s)$ is normally distributed with mean=0 and variance= $t-s$
- By (3.2.13) ($\varphi(u) = e^{\frac{1}{2}u^2t}$), $E[\exp\{\sigma(W(t) - W(s))\}] = \exp\left\{\frac{1}{2}\sigma^2(t - s)\right\}$

$$\therefore \exp\left\{\sigma W(s) - \frac{1}{2}\sigma^2 t\right\} \cdot E[\exp\{\sigma(W(t) - W(s))\}]$$

$$= \exp\left\{\sigma W(s) - \frac{1}{2}\sigma^2 t\right\} \cdot \exp\left\{\frac{1}{2}\sigma^2(t - s)\right\}$$

$$= \exp\left\{\sigma W(s) - \frac{1}{2}\sigma^2 s\right\} = Z(s)$$

first passage time

- Let m be a real number, and define the first passage time to level m

$$\tau_m = \min\{t \geq 0; W(t) = m\}.$$

- This is the first time the Brownian motion W reaches the level m .
- If the Brownian motion never reaches the level m , we set $\tau_m = \infty$
- A martingale that is stopped (“frozen”) at a stopping time is still a martingale and thus must have constant expectation. (more detail: Theorem 4.3.2 of Volume I)
- Because of this fact,

$$1 = Z(0) = EZ(t \wedge \tau_m) = E\left[\exp\left\{\sigma W(t \wedge \tau_m) - \frac{1}{2}\sigma^2(t \wedge \tau_m)\right\}\right]$$

where the notation $t \wedge \tau_m$ denotes the minimum of t and τ_m

first passage time

- We assume that $\sigma > 0$ and $m > 0$. In this case, the Brownian motion is always at or below level m for $t \leq \tau_m$ and so

$$0 \leq \exp\{\sigma W(t \wedge \tau_m)\} \leq e^{\sigma m}$$

$$\exp\left\{\sigma W(t \wedge \tau_m) - \frac{1}{2} \sigma^2 (t \wedge \tau_m)\right\}$$

- If $\tau_m < \infty$, the term

$$\exp\left\{-\frac{1}{2} \sigma^2 (t \wedge \tau_m)\right\} = \exp\left\{-\frac{1}{2} \sigma^2 \tau_m\right\}$$

- If $\tau_m = \infty$, the term

$$\exp\left\{-\frac{1}{2} \sigma^2 (t \wedge \tau_m)\right\} = \exp\left\{-\frac{1}{2} \sigma^2 t\right\}$$

and as $t \rightarrow \infty$, this converge to zero.

- We capture these two cases by writing

$$\lim_{t \rightarrow \infty} \exp\left\{-\frac{1}{2} \sigma^2 (t \wedge \tau_m)\right\} = I_{\{\tau_m < \infty\}} \exp\left\{-\frac{1}{2} \sigma^2 \tau_m\right\}$$

$$\exp\left\{\sigma W(t \wedge \tau_m) - \frac{1}{2}\sigma^2(t \wedge \tau_m)\right\}$$

- If $\tau_m < \infty$, then

$$\exp\{\sigma W(t \wedge \tau_m)\} = \exp\{\sigma W(\tau_m)\} = e^{\sigma m}$$

where t becomes large enough.

- If $\tau_m = \infty$, then we do not know what happens to $\exp\{\sigma W(t \wedge \tau_m)\}$ as $t \rightarrow \infty$, but we at least know that this term is bounded because of

$$0 \leq \exp\{\sigma W(t \wedge \tau_m)\} \leq e^{\sigma m}$$

- That is enough to ensure that

$$\lim_{t \rightarrow \infty} \left(\exp\{\sigma W(t \wedge \tau_m)\} \cdot \exp\left\{-\frac{1}{2}\sigma^2\tau_m\right\} \right) = 0$$

first passage time

- In conclusion, we have

$$\lim_{t \rightarrow \infty} \exp\left\{\sigma W(t \wedge \tau_m) - \frac{1}{2}\sigma^2(t \wedge \tau_m)\right\} = I_{\{\tau_m < \infty\}} \exp\left\{\sigma m - \frac{1}{2}\sigma^2\tau_m\right\}$$

- We take limit

$$1 = Z(0) = EZ(t \wedge \tau_m) = E\left[\exp\left\{\sigma W(t \wedge \tau_m) - \frac{1}{2}\sigma^2(t \wedge \tau_m)\right\}\right]$$

and obtain (interchange of limit and expectation, by the Dominated Convergence Theorem, Theorem 1.4.9)

$$1 = E\left[\lim_{t \rightarrow \infty} \left\{\exp\left\{\sigma W(t \wedge \tau_m) - \frac{1}{2}\sigma^2(t \wedge \tau_m)\right\}\right\}\right] = E\left[I_{\{\tau_m < \infty\}} \exp\left\{\sigma m - \frac{1}{2}\sigma^2\tau_m\right\}\right]$$

or, equivalently,

$$E\left[I_{\{\tau_m < \infty\}} \exp\left\{-\frac{1}{2}\sigma^2\tau_m\right\}\right] = e^{-\sigma m}$$

hold when m and σ are positive.

first passage time

- Since it holds for every positive σ , we can take the limit on both sides as $\sigma \downarrow 0$.
- This yields $E[I_{\{\tau_m < \infty\}}] = 1$ (use the Monotone Convergence Theorem, Theorem 1.4.5) or, equivalently,

$$P\{\tau_m < \infty\} = 1$$

- Because τ_m is finite with probability one (almost surely), we may drop the indicator of this event

$$E\left[I_{\{\tau_m < \infty\}} \exp\left\{-\frac{1}{2}\sigma^2\tau_m\right\}\right] = e^{-\sigma m}$$

and obtain

$$E\left[\exp\left\{-\frac{1}{2}\sigma^2\tau_m\right\}\right] = e^{-\sigma m}$$

Theorem 3.6.2

- For $m \in \mathbb{R}$, the first passage time of Brownian motion to level m is finite almost surely, and the Laplace transform of its distribution is given by $Ee^{-\alpha\tau_m} = e^{-|m|\sqrt{2\alpha}}$ for all $\alpha > 0$

Proof of Theorem 3.6.2

- $E\left[\exp\left\{-\frac{1}{2}\sigma^2\tau_m\right\}\right] = e^{-\sigma m} \quad (3.6.8)$
- When m is positive. Set $\sigma = \sqrt{2\alpha}$, so that $\frac{1}{2}\sigma^2 = \alpha$
- If m is negative, then because Brownian motion is symmetric, the first passage times τ_m and $\tau_{|m|}$ have same distribution.
- Equation
$$Ee^{-\alpha\tau_m} = e^{-|m|\sqrt{2\alpha}} \text{ for all } \alpha > 0$$
for negative m follows.

Remark 3.6.3

- Differentiation of

$$Ee^{-\alpha\tau_m} = e^{-|m|\sqrt{2\alpha}} \text{ for all } \alpha > 0$$

with respect to α results in

$$E[\tau_m e^{-\alpha\tau_m}] = \frac{|m|}{\sqrt{2\alpha}} e^{-|m|\sqrt{2\alpha}} \text{ for all } \alpha > 0.$$

- Letting $\alpha \downarrow 0$, we obtain $E\tau_m = \infty$ so long as $m \neq 0$